

# Alessandro Colace

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## PROFESSIONAL SUMMARY

Senior production engineer with 15+ years shipping backend, platform, and data systems end-to-end - continuously active in tech since high school (first .NET role at 18, never stopped between jobs, side projects, hackathons, and open-source), plus a 2020-2022 founder-like detour at Zenkis (Kyiv) as Operations Manager of a 50-person shoe manufacturer - full P&L, proprietary military + casual brands, turnaround in 12 months until the war. Currently Forward Deployed Engineer at Avatar Investments ([Palantir](#) Official Partner) where I built Avatar Hub - the firm's internal SaaS spanning ATS, AI Studio (Anthropic Claude), Partnership CRM, IAM, AI Maturity Index, Events, Timesheet, public website (EN/IT) and more - on FastAPI · React 19/Vite/TS · Neon Postgres · Microsoft Graph · Azure AD SSO. Earlier: designed Reply's data-lake architecture with first batch + real-time **Kafka** ingestion procedures and led 2 teams of ~10 (Bologna, 2017-2019). **Engineered 70% GCP cost reduction worth €500K+/year** via re-architected query patterns and right-sized Dataflow / BigQuery slot reservations; built a harmonization framework unifying **30+ data providers** for a global retail/manufacturing client. Hands-on across **Anthropic Claude · OpenAI · OpenRouter** in production. Four GCP Professional certifications. **Portfolio:** [avatar-investments.com](#) · [coalesce.coach](#) · [hedge-fund-tracker](#).

## CORE COMPETENCIES

Data Engineering · Lakehouse (Iceberg · Delta)

Python · FastAPI · ETL/ELT

Cloud (GCP 4x cert · Azure · AWS)

Team Leadership (4-20 engineers)

CI/CD · Docker · K8s · IaC

Insurance / FinTech delivery (Unipol, Telepass)

## WORK EXPERIENCE

### Avatar Investments

Jan 2026 - Present

**Forward Deployed Engineer** - Milan, Italy

Avatar Investments is a **Palantir Official Partner** delivering enterprise AI transformations, with offices in Milan, Istanbul, and Miami.

- **Built Avatar Hub**, the firm's unified internal SaaS platform - used daily across ATS / hiring with public Careers application + Chrome MV3 LinkedIn-import extension, AI Studio (Anthropic Claude chat + deck export + Deep Research), Partnership CRM, IAM with role-based module access, AI Maturity Index public lead-gen, Events with NDA SES flow + SharePoint Graph proxy, Meetings/Polls + Emails (Microsoft Graph), Timesheet, public website (EN/IT), Onboarding tours, Certifications, WhatsApp sidecar (Baileys/Express/WS). Stack: FastAPI · Uvicorn · SQLAlchemy 2.x · APScheduler · React 19 · Vite 8 · TypeScript 6 · Zod · Zustand · Tailwind v4 · Vitest · Playwright cross-browser · single shared Neon Postgres · Azure AD SSO + MSAL · Cloudflare CSP/HSTS · Render. Pre-push enforced: tsc + vitest + pytest. GDPR-rigorous: PII masking on public-unauth logs, public Pydantic extra='forbid', SSRF guard, NDA ROPA + Art. 6(1)(f) balancing. **Designed AI-coded-onboarding ready:** extensive CLAUDE.md + per-domain conventions + canonical-files index so any new contributor (or LLM-assisted agent) is productive from day zero.
- Lead enterprise AI delivery on **Palantir Foundry & AIP** (AI Ontology, Mission Engineering / Execution), translating customer business needs into deployed agents and governed data products.
- Author / contributor to Avatar's **FDE methodology playbook** - the firm-wide engagement framework (modeled on the [8090](#) direction, structured as an [Obsidian](#)-style knowledge vault) covering AI Assessment, Business Development, Commercial Model, Delivery Framework, vertical-specific guides (manufacturing, retail, education, F&B), and Palantir-specific transformation playbooks.

### COalesCE

Jun 2022 - Jan 2026

**Cloud Data & AI Architect** - Milan, Italy

Independent cloud-data consulting practice across Reply group (Go Reply, Cluster Reply), The Information Lab, and Nimbus Intelligence. Selected clients: **Telepass, Campari, Ferragamo, IEO, Unipol**.

- Led data teams of 4 to 8 people, implementing Scrum to achieve **100% on-time delivery** for 3 major projects, eliminating missed deadlines.
- Architected a data harmonization framework for a global retail/manufacturing client, unifying data from 30+ providers by resolving inconsistencies in currency, timezones, and product units.
- Developed scalable and cost-efficient cloud data platforms; achieved a **70% cost reduction on GCP saving €500K+ annually**.
- Partnered with senior stakeholders to design a Unified Customer View (UCV) for ~8M customers and a complex ARPU model, integrating diverse revenue and promotions streams.
- Established GDPR-compliant Data Governance framework, integrating Data Catalog and dbt for automated Data Quality checks, improving accuracy, consistency, and timeliness KQIs by ~30%.

### Zenkis (Shoes Manufacturing)

Jan 2020 - May 2022

**Operations Manager** - Kyiv, Ukraine

50-person shoe manufacturing company. **Founder-like scope:** full P&L ownership across every business function (operations, sales, HR, marketing, production, procurement, R&D) plus entrepreneurial **proprietary brand-building (military + casual lines)**.

- Owned end-to-end every business function with full P&L responsibility on a 50-person manufacturing operation.
- Engineered a turnaround from **€10K/month losses to profitability inside 12 months** via **Lean Manufacturing**, waste reduction, and demand-driven production scheduling.
- Built proprietary shoe brands (military + casual) from product design through retail go-to-market. War (Feb 2022) forced the pivot back to software.

Data Reply

Jan 2017 - Dec 2019

Senior Big Data Architect - Bologna, Italy

- Actively supported the opening and scale-up of Reply's Bologna office: personally sourced 5+ engineers and secured a new retail client generating **€1M+/year** for Reply Holding (€2B+ revenue, 15k+ employees).
- Led 2 teams of ~10 people each (Data Viz and Integration + Data Engineering).
- **Core client: Unipol** (~80% of the Reply tenure) - end-to-end data work plus C-level Tableau KPI dashboards unifying siloed departments. Designed the data-lake architecture (raw / processing / service / insights layers) on Cloudera Hadoop with first batch + real-time **Kafka** ingestion procedures. Additional engagements: YOOX Net-a-Porter (data + Tableau).

Iconsulting

Oct 2015 - Dec 2016

Business Intelligence Consultant - Casalecchio di Reno (BO), Italy

- Implemented a proactive alerting system for Big Data Analytics within a Tier-1 telco context (high-impact Master's thesis); maintained proprietary Location Intelligence framework on MongoDB / Angular / Leaflet / Kafka / Node.

Various Roles

Jan 2010 - Oct 2015

Tutoring, Java, Sysadmin, E-Commerce (post-Sitech, parallel to studies) - Bologna & Ferrara, Italy

- **University of Bologna - Computer Engineering Tutor** (Nov 2014 - Oct 2015, part-time). 🏆 **H-ACK TRAVEL 1st place** (Apr 2014, H-FARM Roncade) - our PLAY-n-PLANE app for group travel itineraries won the 24-hour hackathon for Bravofly.
- **Unimatica S.p.A. - Java Developer** (Jul - Dec 2013): backend components for public-sector digital signature workflows; issued and revoked Infocert and Actalis certificates.
- **StarEmilia Srl (Mercedes-Benz dealership) - Solo Sysadmin** (Sep 2011 - Feb 2012, part-time): full IT stack ownership.
- **Rian Next Abbigliamento - E-Commerce Manager** (Jan - Dec 2010, part-time): eBay shop and online retailing for a family-owned fashion clothing store.

Sitech Srl (via Value Team, later NTT Data)

Sep 2008 - Sep 2009

.NET Developer - Bologna, Italy

- **First full-time permanent role at age 18**, straight out of high school (ITIS Belluzzi). Recruited by my former CS teacher after I **exposed a security breach in my school's IT system that drew national media attention**; treated as senior developer from day one. Built enterprise apps on ASP.NET MVC (C#), MSSQL, Crystal Reports for an integrated **rail traffic control system**.

PROJECTS

Hedge Fund Tracker

**27+ GitHub stars.** Full-stack open-source tool tracking institutional portfolios via SEC 13F/13D/Form 4 filings with real-time AI analysis. **Backend:** Python 3.13 · FastAPI · uvicorn. **Frontend:** React 19 · Vite · TypeScript · Tailwind · TanStack Query · Radix UI. **AI providers integrated:** Google Gemini, OpenAI, Groq, HuggingFace, OpenRouter, GitHub Models. **Deploy:** Docker · GitHub Pages · GitHub Actions CI/CD for daily SEC EDGAR ingestion.

EDUCATION

Master's Degree in Computer Engineering - University of Bologna

Dec 2013 - Jul 2016

**110/110 Summa Cum Laude.** Thesis: *Design and Implementation of a Proactive Alerting System inside a Big Data Analytics platform.*

CERTIFICATIONS

|                                                                    |      |
|--------------------------------------------------------------------|------|
| Six Sigma Yellow Belt (SSYB) - Council for Six Sigma Certification | 2025 |
| Machine Learning Engineer (PMLE) - Google Cloud                    | 2024 |
| Cloud Network Engineer (PCNE) - Google Cloud                       | 2024 |
| Cloud DevOps Engineer (PCDOE) - Google Cloud                       | 2023 |
| Cloud Database Engineer (PCDE) - Google Cloud                      | 2023 |
| Tableau Desktop Specialist (TDS-C01) - Tableau                     | 2018 |

SKILLS

**AI & Agentic:** Anthropic Claude · OpenAI · Google Gemini · Groq · HuggingFace · OpenRouter · GitHub Models · LangChain · Vertex AI · Palantir AIP & Foundry · RAG · MCP · Multi-Agent · prompt engineering

**Backend & Platform:** FastAPI · Node.js · Java EE · ASP.NET MVC · REST · OAuth / OIDC · Microsoft Graph · Azure AD / Entra ID / MSAL · Multi-tenant Postgres · RBAC · audit logging · observable systems

**Data & Lakehouse:** BigQuery · Snowflake · Databricks · dbt · Apache Airflow · Spark / PySpark · Kafka · Iceberg / Delta / Parquet · Medallion architecture · Data Contracts · SLO-based Data Quality

**Cloud & Infra:** GCP (4x Professional cert) · AWS · Azure · Terraform · Docker · Kubernetes · GitHub Actions · Playwright · CI/CD

**Programming:** Python · TypeScript · JavaScript · SQL · Java · PySpark · Scala · Shell · C# / .NET · React 19 · Vite · Tailwind

**Languages:** Italian (native) · English (C1, fluent) · Russian (base - 2+ years in Ukraine)